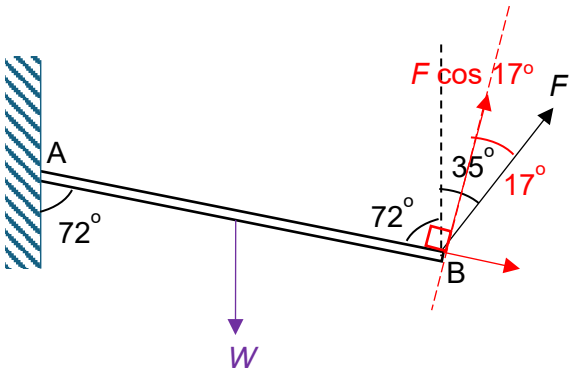
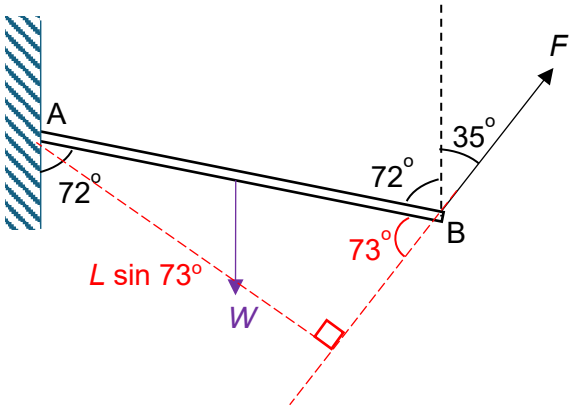
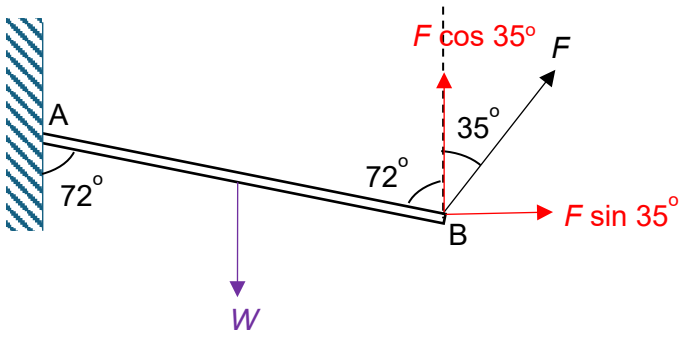
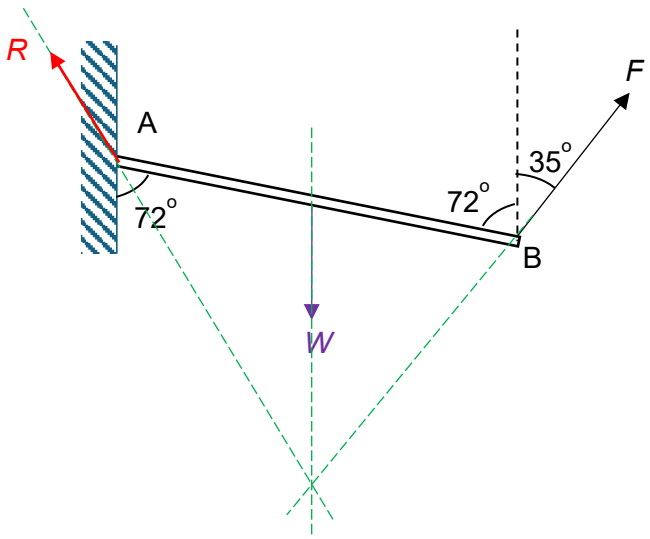
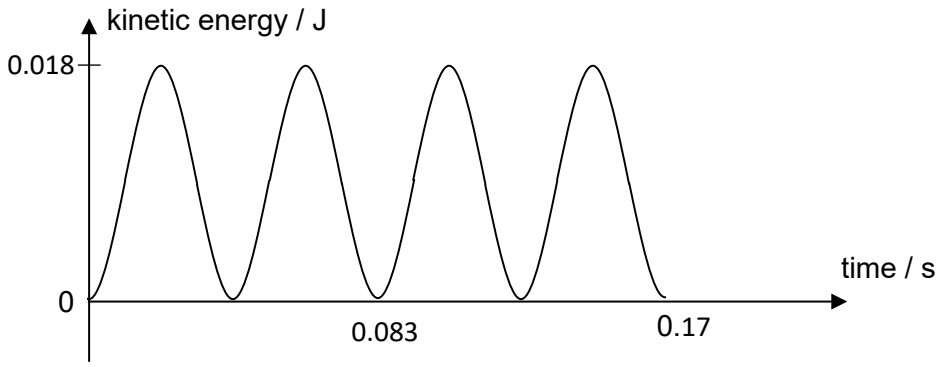
Paper 3

Question 1		
(a)	Let the length AB be *L*.  72°+35°-90°=17° Taking moments about A: Anti-clockwise moment due to component of *F* perpendicular to AB = Clockwise moment due to weight $F\cos 17^o\left(L\right)=W\left(0.5L\sin 72^o\right)$ $F=5.967=6.0$ N	**B1** **A0**
	Alternative:  Taking moments about A: Anti-clockwise moment due to *F* = Clockwise moment due to weight $F\left(L\sin 73^o\right)=W\left(0.5L\sin 72^o\right)$ $F=5.967=6.0$ N	

	Alternative:  Taking moments about A: Anti-clockwise moment due to F cos 35° + anti-clockwise moment due to F sin 35° = Clockwise moment due to weight $F\cos 35^o\left(L\sin 72^o\right)+F\sin 35^o\left(L\cos 72^o\right)=W\left(0.5L\sin 72^o\right)$ $F=5.967=6.0$ N	
(b)(i)	 Note to marker: Ignore length of arrow, working lines are expected	**A1**
(b)(ii)	For equilibrium, $\sum F_x=0$ $R_x=F_x=6.0\sin 35^o$ $\sum F_y=0$ $R_y+F_y=W$ $R_y=12-6.0\cos 35^o$ $R=\sqrt{R_x^2+R_y^2}=7.89=7.9$ N Note to marker: Substituting unrounded F gives the same final answer (2 s.f.).	**B1** **B1** **A1**

Question 2		
(a)(i)	T = 24 x 3600 = 86400 s	**A1**
(a)(ii)	Gravitational force provides centripetal force $\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$ $r^3 = \frac{GMT^2}{4\pi^2}$ $r = \sqrt[3]{\frac{(6.67\times10^{-11})(5.97\times10^{24})(86400)^2}{4\pi^2}} = 4.22\times10^7$ m	**B1** **A1**
(b)(i)	$\Delta\phi = -\frac{GM}{r_{far}} - (-\frac{GM}{r_{near}}) = GM(\frac{1}{r_{near}} - \frac{1}{r_{far}})$ $\Delta\phi = (6.67\times10^{-11})(5.97\times10^{24})(\frac{1}{6.87\times10^6} - \frac{1}{4.62\times10^7}) = 4.93\times10^7$ J kg^{-1}	**B1** **A1**
(b)(ii)	Loss in KE = Gain in GPE = $m\Delta\phi$ = $1500(4.93\times10^7) = 7.40\times10^{10}$ J Change in KE from closest to furthest = -7.40 x 10^{10} J Note to marker: Deduct 1 mark if a positive answer is given.	**M1** **A1**
(b)(iii)	The satellite moves slowest at its furthest point (due to minimum kinetic energy), meaning it spends the greatest proportion of its orbital time / lingers over the polar region.	**A1**

Question 3		
(a)	$\omega = 2\pi f = 2\pi(12) = 75.4 = 75 \text{ rad s}^{-1}$	**A1**
(b)(i)	$a_0 = \omega^2 x_0 = (75)^2 (0.016)$ $= 90 \text{ m s}^{-2}$	**C1** **A1**
(b)(ii)	Weight is in same direction as contact force at top, opposite to contact force at bottom. Same net force at both positions, hence the contact force at bottom is larger.	**B1** **B1**
(c)	$\frac{1}{2}mv_0^2 = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2}(0.025)(75)^2(0.016)^2 = 0.018 \text{ J}$ $T = \frac{1}{f} = \frac{1}{12} = 0.08333 \text{ s}$  **B1** – correct number of cycles **B1** – correct shape	**A1**

Question 4

(a)

There is an increase in separation hence an increase in the molecular potential energy. **B1**

Temperature remains constant so the microscopic kinetic energy remains constant. **B1**

(b)(i)

$n = pV / RT$

$= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273.15 + 17)]$

$= 12$ mol **A1**

Marker's comments

(b)(ii)

change	work done on gas/kJ	thermal energy supplied to gas /kJ	increase in internal energy of gas/ kJ
AB	19.2	–48.0	–28.8
BC	0	28.8	28.8
CA	–31.6	31.6	0

Work done AB = $1.2 \times 10^5 \times (0.24 – 0.08) = 19200$ J = 19.2 kJ

Work done BC = 0 (since no change in volume)

Work done CA= –31.6 (value given in question, students to take note that the work done is negative since CA is an expansion process)

[1 mark for work done]

CA: increase in internal energy = 0 (since no change in temperature) and thermal energy supplied = 31.6 (by first law of thermodynamics)

[1 mark for CA]

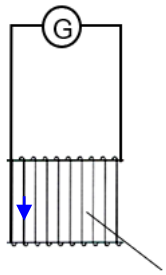
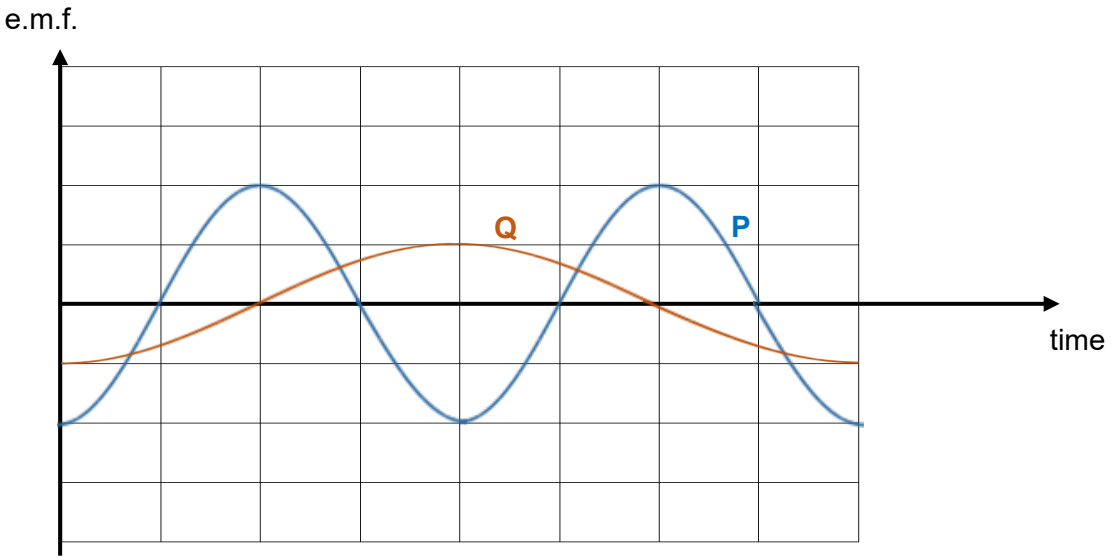
Increase in internal energy of AB = -28.8, increase in internal energy of BC = 28.8

Change in internal energy (AB + BC + CA) = 0

[1 mark for increase internal energy and thermal energy of BC]

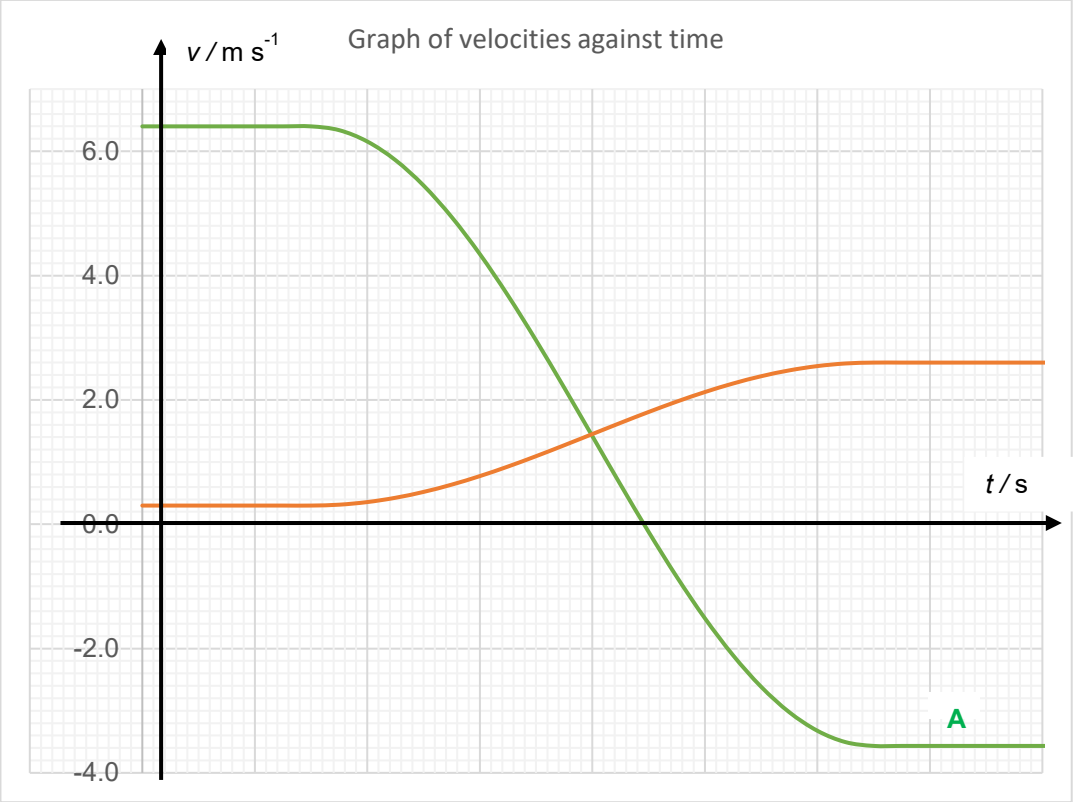
Question 5		
(a)	$I = \frac{E}{R} \Rightarrow \frac{\Delta I}{I} = \frac{\Delta R}{R}$ (E is a constant.) $\Delta I = 5\% \times (\frac{E}{R}) = 0.05 \times \frac{9.0\ \text{V}}{10\ \Omega} = 0.045$ A = 0.05 A (to 1 s.f.) <u>Alternative:</u> Max. value of R = 1.05 x 10 Variation in I = $\frac{9.0}{10} - \frac{9.0}{1.05 \times 10} = 0.0428$ A ~ 0.05 A <u>Alternative:</u> • min value of R - variation will be 0.0474 A, or • (max – min) /2 (variation will be 0.0451 A)	**B1** **A1**
(b)(i)	Any correct explanation, e.g.: • The actual results differs from the predicted results by more than 5% (up to approximately 40%). • If it was a random variation, then the actual results should deviate in such a way that it sometimes higher and sometimes lower than the predicted results (but still producing a straight line trend). • Actual results is always lower than the predicted results suggesting a systematic error is present.	**A1**
(b)(ii)	As more resistance is added to the parallel network, the **effective resistance of the network of resistors reduces**. The external circuit and the internal resistance of the battery form **a potential divider**, so a **smaller fraction of the emf is across the external circuit / the terminal p.d. decreases**, causing the measured current to be increasingly lower than the predicted current. <u>Alternative:</u> As more resistors are connected in parallel, **the effective resistance decreases and the current drawn increases.** Therefore, the **potential drop across the internal resistance increases**, so the **terminal p.d. across the external circuit decreases** and the measured current becomes increasingly lower than predicted.	**B1** **B1** **B1** **B1**
(c)	$E = IR + Ir \Rightarrow \frac{1}{I} = (\frac{1}{E})R + \frac{r}{E}$ $\Rightarrow \frac{r}{E} = r$(gradient) = vertical intercept Using (1.6, 0.28) and (9.0, 1.12), Gradient of graph = $\frac{(1.12) - (0.28)}{(9.0) - (1.6)} = 0.1135$	**B1** **B1**

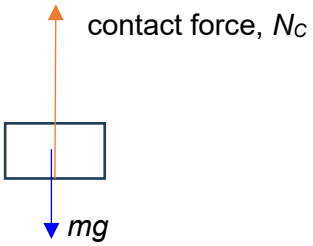
	Note to marker: • Accept anything from 0.10 to 0.12. • Deduct this mark if working is not shown for gradient. $r = \frac{\text{vertical intercept}}{\text{gradient}} = \frac{(0.10)}{0.1135} = 0.88\ \Omega$ Accept solution if student substitutes point(s) from best-fit line and solve simultaneous equations to get the final internal resistance e.g. using two points on BFL or a point with gradient. [-1] if the student uses data point does not lie on the BFL or use 9.0 V as the emf.	**A1**

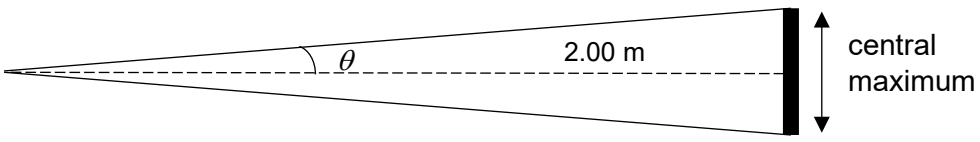
Question 6		
(a)(i)	The e.m.f. induced in a conductor is proportional to the rate of change of magnetic flux linkage. Magnetic flux density generated by solenoid is decreasing (as the current flowing through the solenoid is decreasing). Hence, magnetic flux linkage in small coil is decreasing, so e.m.f. is induced.	**B1** **B1**
(a)(ii)	By Lenz's law, the magnetic field set up in the small coil due to the induced current will be in **the same (rightward) direction as the magnetic field generated by the solenoid** to **oppose the decrease in the flux linkage.** Apply right hand grip rule to determine direction of induced current. Direction of induced current marked correctly on diagram (Induced current should be drawn on a wire and pointing down).  	**B1** **A1**
(b)(i)	 Cosine graph plotted (positive or negative accepted) $I = I_0 \sin\omega t + c$ where $c > 0$ Since $B \propto I$, $\varepsilon = -\frac{d(NBA)}{dt} = -NA\frac{dB}{dt}$ Hence, $\varepsilon \propto -\cos\omega t$	**A1**
(b)(ii)	Peak value halved Period doubled	**A1** **A1**

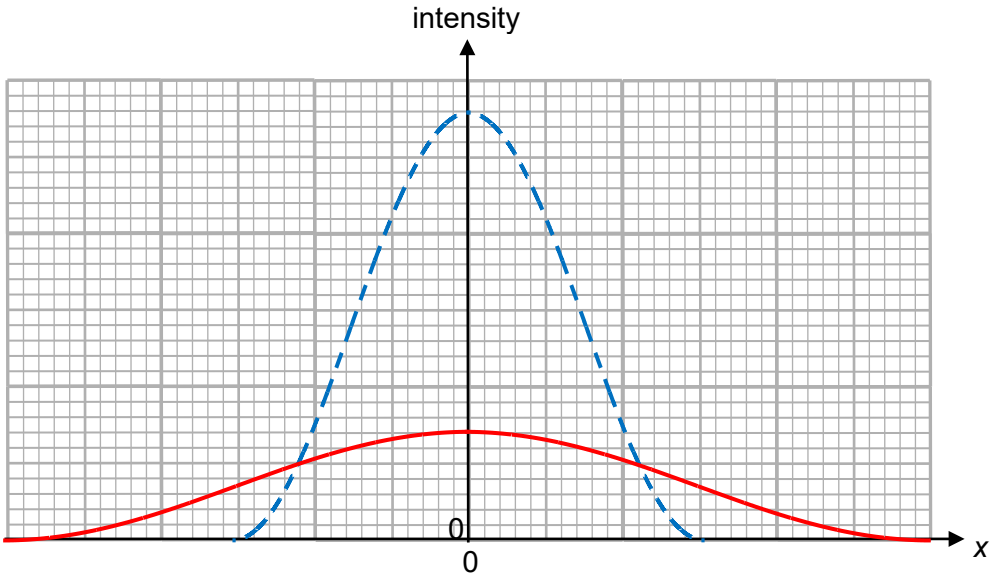
Question 7		
(a)	Below the threshold frequency, no electrons are emitted (no matter the intensity). Light interacts with the electrons as a photon (particle) of energy h*f*. h*f* must be greater than the metal's work function to liberate an electron.	**B1** **B1**
(b)	There is significant diffraction when the wavelength is comparable to or larger than the object's dimensions. Visible light has wavelength of 380 nm to 750 nm, while an electron (beam's kinetic energy could be adjusted to give it a) wavelength smaller than 140 nm.	**B1** **B1**
(c)	According to de Broglie's relation, photons have momentum $p = \frac{h}{\lambda}$. They experience a change in momentum when they bounce off the sail. The force that the sail exerts on them is equal to the force they exert on the sail.	**B1** **B1**

Question 8		
(a)	Uranium, barium, and krypton are located towards the **right-hand side of the peak** of the binding energy curve shown in Figure 8.1. Barium and krypton have a **higher binding energy per nucleon** than uranium, as uranium has a higher nucleon number Consequently, the **total binding energy** of the resulting barium and krypton nuclei is greater than the total binding energy of the initial uranium nucleus. This means that **more energy is released** when the barium and krypton nuclei are formed from free nucleons than the **energy required to separate** the uranium nucleus into its constituent nucleons. <u>Alternative:</u> Consequently, the **total binding energy** of the resulting barium and krypton nuclei is greater than the binding energy of the initial uranium nucleus. This means that (the total mass defect decreases after the reaction and) the **products have smaller total mass**. Hence, this nuclear fission process results in a net release of energy.	**B1** **B1**
(b)(i)	[92 ×mass of proton + 143 × mass of neutron − mass of uranium nucleus]c^2 = BE of uranium $(92\times1.00814\text{u}+143\times100898\text{u}-\text{mass}_{\text{Uranium}})c^2=235\times7.5909\times1.6\times10^{-13}$ $92\times1.00814\text{u}+143\times1.00898\text{u}-\text{mass}_{\text{Uranuim}}=1.91043\text{u}$ $\text{mass}_{\text{uranium}}$ = 235.12259u	**M1** **M1** **A0**
(b)(ii)	Energy released = (mass of reactants − mass of products)c^2 = $(235.12259-140.91441-91.92615-2\times1.00898)\text{u}c^2$ = 3.95×10^{-11} J	**C1** **A1**
(iii)	The initial colliding neutron has kinetic energy hence by conservation of mass energy, the total kinetic energy of the products will need to include the kinetic energy of the colliding neutron.	**B1**

Question 9		
(a)	The **total linear momentum** of a system **remains unchanged** if **no net external force** acts on the system	**B1** **B1**
(b)(i)	Applying conservation of linear momentum: $m_A u_A + m_B u_B = m_A v_A + m_B v_B$ $(1.2)(6.40) + (5.2)(0.30) = 1.2v_A + 5.2v_B$ Since collision is elastic, relative speed of approach = relative speed of separation $u_A - u_B = v_B - v_A$ $v_A = v_B - 6.10$ Solving, $1.2v_A + 5.2(v_A + 6.10) = 9.24$ or $1.2(v_B - 6.10) + 5.2v_B = 9.24$ $v_B = 2.5875 = 2.6 \text{ m s}^{-1}$	**B1** **B1** **B1** **A0**
(b)(ii)	 From **(i):** $v_A = 2.6 - 6.1 = -3.5 \text{ m s}^{-1}$ • Correct flat lines for where relevant • A smooth and downward sloping curve in between One mark is deducted for any mistake made	**A1** **A2**
(b)(iii)	Draw a tangent at the time on the curve of variation of velocity with time of trolley B. Determine its gradient and multiply it with the mass of trolley B to arrive at the force on trolley B.	**B1** **B1**

(b)(iv)	The total kinetic energy decreases upon collision and reaches a (non-zero) minimum value. After which, total kinetic energy increases and returns to its initial total kinetic energy before collision.	**B1** **B1**
(b)(v)	At maximum compression, the two trolleys have the same speed. Applying conservation of linear momentum: $m_A u_A + m_B u_B = (m_A + m_B)v_C$ $(1.2)(6.40) + (5.2)(0.30) = (1.2 + 5.2)v_C$ $v_C = 1.444 \text{ m s}^{-1}$ Applying conservation of energy Gain in E.P.E. = Loss in K.E. $= (0.5)(1.2)(6.40^2) + (0.5)(5.2)(0.30^2) - (0.5)(1.2 + 5.2)v_c^2$ $= 18.14$ $= 18.1 \text{ J}$ EPE in each spring = 9.1 J	**B1** **B1** **A1**
(c)(i)	v_Y is the vertical velocity of trollley before colliding with the ground, $v_y^2 = 2(9.81)(10.0)$ $v = 14.0 \text{ ms}^{-1}$ Average vertical force = $= \frac{m\Delta v}{\Delta t} = \frac{5.2(14.0 - 0)}{0.100} = 728 \text{ N}$ (upwards) Applying Newton's second law to the vertical direction: $N_C - mg = 728$ $N_C = 728 + 5.2(9.81) = 779 \text{ N}$  Note to marker: Award max 2 marks for students who ignore weight in their analysis.	**B1** **B1** **B1** **A1**
(c)(ii)	Impulse is the change in momentum, so impulse (in the vertical direction) is larger. (*Impulse in the horizontal direction depends on the angle of rebound. Students can discuss this but as long as their deduction makes sense.*)	**B1**

Question 10		
(a)	Coherent light waves are electromagnetic waves that have a **constant phase difference** between them.	**B1**
(b)(i)	Bright fringes are formed by the **constructive interference** of the coherent waves emerging from the two slits arriving on the screen **in phase**. [or with a constant phase difference of m x 2π rad where m: order (m= 0,1, 2 3…).] Dark fringes are formed by the **destructive interference** of the coherent waves arriving on the screen **out of phase.** [or with a constant phase difference of $(m-\frac{1}{2})\pi$ rad.] Note to marker: will only accept phase difference rather than path difference as phase difference is the fundamental condition for the outcome of the interference	**B1** **B1**
(b)(ii)	Using the fringe separation for double slit interference: fringe separation, $y = \frac{\lambda L}{d}$ $= \frac{(633 \times 10^{-9})(2.00)}{0.250 \times 10^{-3}}$ $= 5.064 \times 10^{-3}$ m Distance between the central bright fringe and the 2nd order dark fringe = 1.5y = 1.5 x 5.064 x 10^{-3} = 7.596 x 10^{-3} m = 7.60 mm	**C1** **A1**
(b)(iii)	The double-slit interference fringes are modulated by the single-slit diffraction envelope resulted from the appreciable width of each slit. As the single-slit diffraction envelope has its greatest intensity at the centre, the maximum observed at O is of a greater intensity than any other observable maximum.	**B1** **B1**
(c)(i)	The angle θ of first minimum for the single slit is given by: $\sin\theta = \frac{\lambda}{b}$ $\sin\theta = \frac{633\times10^{-9}}{0.080\times10^{-3}} = 0.0079125$ For small angle θ, $\sin\theta \approx \tan\theta$  $\therefore$ width of the central maximum = 2 x 2.00 (tan θ) = 2 x 2.00 x 0.0079125 = 0.03165 m = 31.7 mm	**B1** **B1** **A0**

	Note to marker: Students who do not use small angle approximation will have the same value.	
(c)(ii)	 Marking point: • Peak intensity is smaller. (1/4 of original intensity not required) • Width doubles	**A1** **A1**
(c)(iii)	Using the Rayleigh criterion for the resolving power of a single aperture $\theta_m \approx \frac{\lambda}{b}$ Using small angle approximation for the angle subtended by the two sources at the aperture, θ_m (in rad) $\approx \frac{s}{D}$ $\Rightarrow \frac{s}{D} = \frac{\lambda}{b}$ $\therefore D = \frac{s \times b}{\lambda} = \frac{(5.0 \times 10^{-3})(0.080 \times 10^{-3})}{633 \times 10^{-9}}$ $= 0.632$ $= 0.63$ m	**B1** **C1** **A1**
(d)(i)	Maximum angle of diffraction $\theta = \tan^{-1}\left(\frac{3.00}{2.00}\right) = 56.31^o$ From the grating equation $d \sin\theta = n\lambda$, $n = \frac{\sin\theta}{N\lambda}$ where N is the number of slits per metre $n = \frac{\sin 56.31^o}{(500 \times 10^3)(633 \times 10^{-9})} = 2.6$ Hence maximum observable order = 2 Maximum number of observable bright fringes = 2(2) + 1 = 5	**B1** **B1** **C1** **A1**

(d)(ii)	The **first order maximum** shows a **continuous spectrum of colours** (with red furthest from the central maximum and violet closest to the central maximum) while **the central maximum shows a** (narrow) **white line/spot.** From the diffraction equation $d \sin\theta = n\lambda$. For $n = 0$, all the wavelengths overlap, producing the white line/spot in the central maximum. In the first order maximum ($n = 1$), **different wavelengths deviate with different angles** θ with the longer wavelength deviating with the larger angle hence producing the spread of colours in the continuous spectrum.	**B1** **B1**